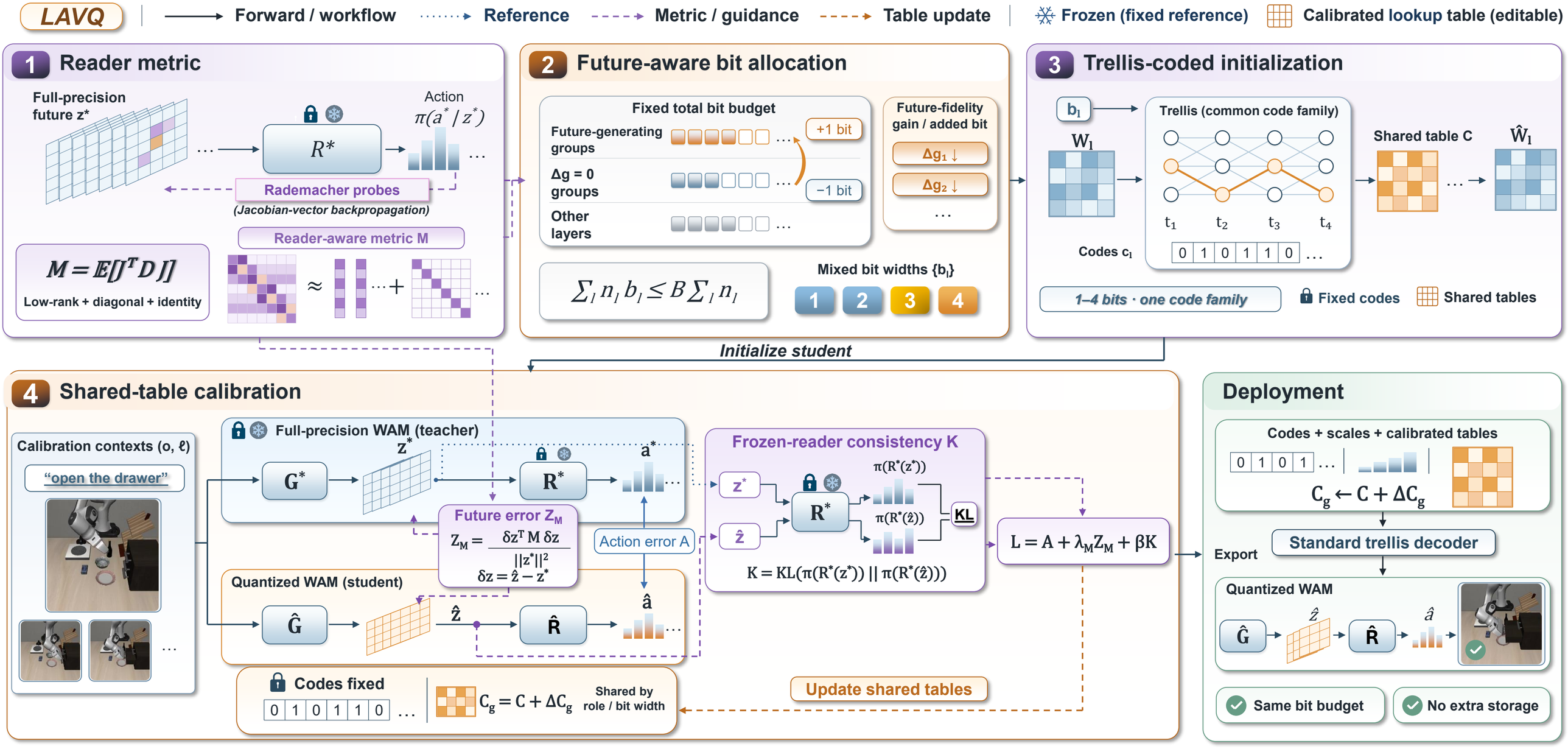

LAVQ
Forward / workflow
Reference
Metric / guidance
Table update
Frozen (fixed reference)
Calibrated lookup table (editable)
1 Reader metric
Full-precision future z*
R*
Action
π(a* | z*)
Rademacher probes
(Jacobian-vector backpropagation)
Reader-aware metric M
M = E[JᵀDJ]
Low-rank + diagonal + identity
2 Future-aware bit allocation
Fixed total bit budget
Future-generating groups
+1 bit
Δg = 0 groups
−1 bit
Other layers
Future-fidelity gain / added bit
Δg₁ ↓
Δg₂ ↓
Σ_l n_l b_l ≤ B Σ_l n_l
Mixed bit widths {b_l}
1
2
3
4
3 Trellis-coded initialization
b_l
W_l
Trellis (common code family)
t₁ t₂ t₃ t₄
Codes c_l
0 1 0 1 1 0
Shared table C
Ŵ_l
1–4 bits · one code family
Fixed codes
Shared tables
Initialize student
4 Shared-table calibration
Calibration contexts (o, ℓ)
"open the drawer"
Full-precision WAM (teacher)
G*
z*
R*
a*
Future error Z_M
Z_M = δzᵀ M δz / ||z*||²
δz = ẑ − z*
Action error A
Frozen-reader consistency K
π(R*(z*))
π(R*(ẑ))
KL
K = KL(π(R*(z*)) || π(R*(ẑ)))
Quantized WAM (student)
Ĝ
ẑ
R̂
â
L = A + λ_M Z_M + βK
Codes fixed
0 1 0 1 1 0
C_g = C + ΔC_g
Shared by role / bit width
Update shared tables
Deployment
Codes + scales + calibrated tables
0 1 0 1
C_g ← C + ΔC_g
Export
Standard trellis decoder
Quantized WAM
Same bit budget
No extra storage